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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 11/09/2005 |
| **PI Name:** | Zderic, Vesna Ph.D. | | |
| **Project Title:** | Contrast Agents for High-Intensity Focused Ultrasound Hemostasis | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Smart Medical Systems Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
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| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 98195 | **Congressional District:** | 7 |
| **Comments:** | | | |
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| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
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| **Flight Program:** | | | |
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| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Crum, Lawrence ( MENTOR/University of Washington ) | | |
| **Grant/Contract No.:** | NCC 9-58-PF00505 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | POSTDOCTORAL FELLOWSHIP<br>(1) the original aims The hypothesis is that the application of ultrasound contrast agents will improve efficacy of HIFU hemostasis of injured solid organs, while maintaining the treatment safety. Specific Aim 1: Determination of efficacy and safety of HIFU hemostasis in the presence of ultrasound contrast agents. Aim 1a: Mechanisms of HIFU hemostasis in the presence of commercially available UCA Optison. Thermal, mechanical and biological mechanisms of hemostasis in the presence of Optison will be investigated for the treatment optimization. Aim 1b: Efficacy and safety of HIFU hemostasis in the presence of Optison. Hemostasis efficacy will be determined, as well as healing of hemostatic liver incisions in rabbits for up to 60 days after the treatment. Specific Aim 2: Development and testing of autologous UCA for HIFU hemostasis. Aim 2a: Development of autologous UCA. UCA consisting of blood, saline and air will be produced, characterized, and optimized for hemostasis application. Aim 2b: Determination of the efficacy of HIFU | | |

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| **Task Description:** | hemostasis in the presence of autologous UCA. HIFU treatment of liver incisions will be performed in the presence of autologous UCA to determine the treatment efficacy in producing hemostasis. (2) the key findings - Thermal measurements showed similar temperatures of the coagulum formed at the injury site treated with HIFU, with or without UCA administration. - Inertial cavitation activity, a signature of mechanical effects, was higher during HIFU application in the presence of UCA. - The coagulum formation was faster as the concentration of administered Optison increased. - Hemostatic seal over injury site achieved with HIFU or HIFU+UCA was more robust as compared to the seal achieved with electrocautery. - Our results showed that UCA can increase the rate of successful bleeding detection. - The levels of systemic blood hemolysis during HIFU therapy increased if UCA were administered before the therapy. - Light microscopy observations showed no difference between the liver tissue treated with HIFU only or HIFU in the presence of UCA. - We developed a method for occlusion of large blood vessels of up to 1 cm in diameter. - Ultrasound-guided phased array was used for occlusion of blood vessels at different depths inside a liver. - We showed the feasibility of ultrasound-guided HIFU treatment of occult injuries located in the posterior surface of the liver. - The presence of bubbles at the HIFU focus prevented the temperature increase at the tissue-air interface in the post-focal region. - Color Doppler HIFU can be used for transcutaneous detection and sealing of puncture wounds in the femoral arteries in a porcine model. (3) the impact of these findings Our findings showed that faster coagulum formation and hemorrhage control of liver incisions was due to the mechanical effects of HIFU application in the presence of UCA. Initial safety experiments showed a potentially higher blood hemolysis after the administration of UCA. The safety of our method will be explored in more detail during long-term survival studies proposed for the next funding year of this project. Bleeding detection experiments showed that UCA can be used to improve the detection and localization of bleeding sites, which would be of significant importance during HIFU treatment of hidden bleedings and eventual extracorporeal HIFU application. (4) the proposed research plan for the coming year: Specific Aim 1b: Efficacy and safety of Optison-enhanced HIFU hemostasis (Months 13-18). A survival study will be performed to observe healing of liver lacerations for up to 60 days after the treatment. A total of 35 adult New Zealand White rabbits will be divided into 2 groups: Optison treatment (n = 21) and control treatment (HIFU only) (n = 14). Ultrasound images, blood and histological samples will be collected immediately (day 0) and on days 1, 3, 7, 14, 28, and 60 after the treatment. Specific Aim 2a: Investigations for development of in-house UCA for HIFU therapy (Months 13-18). The shelf life of Optison of 2 years may be too short for longer space flights, for example to Mars. Therefore, we will pursue the development of contrast agents that could be made in Space. These could potentially consist of blood or blood components, saline, and air or other biologically compatible gases. This work will provide valuable guidelines on eventual development of fully-autologous contrast agents that can be made on site during flight missions. The properties of the developed UCA such as bubble size, concentrations, life time, and echogenicity, will be studied specifically for therapy applications. Specific Aim 2b: Determination of the efficiency of HIFU treatment in the presence of in-house UCA (Months 19-24). HIFU treatment of liver lacerations in rabbit in-vivo will be performed in the presence of in-house UCA. The application of in-house UCA, optimized in Specific Aim 2a, is expected to provide similar values of the hemostasis time reduction as the application of Optison. Specific Aim 2c: Determination of the utility of in-house UCA for improving the bleeding detection using ultrasound imaging (Months 19-24). The in-house UCA will be administered at the distal site after induction of injury in the posterior surface of solid organs in rabbits. Conventional B-mode, Color Doppler, Pulsed Doppler, Harmonic Imaging, and Pulse Inversion Harmonic Imaging (PIHI) will be used to visualize and potentially locate the bleeding site. We expect that our in-house contrast agents will show similar enhancement of bleeding detection as Optison during the first phases of interrogation by ultrasound. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | On Earth, bleeding due to blunt trauma is one of the primary causes of death within an hour after the injury if adequate care is not provided (Sauaia et al. 1995, Franklin et al. 2000). Rapid achievement of hemostasis is necessary to prevent hemorrhagic shock following intra-abdominal injury (Feliciano et al. 2000). However, hemostasis is not an easy task in trauma patients, due to the usual presence of coagulopathy, resulting from hypothermia, acidosis, and dilution of clotting factors (Bellamy 1984). The difficulties associated with treating battlefield and civilian trauma hemorrhage are in being able to detect and localize the hemorrhage site, and to stop the bleeding non-invasively, all within a very short time (of several minutes) after the trauma has occurred. Our group has demonstrated a potential of ultrasound-guided HIFU to both detect a bleeding site and deliver energy to produce hemostasis. Our current NSBRI-funded project addresses the HIFU hemorrhage control from solid organs in the presence of ultrasound contrast agents (UCA). The administration of UCA may facilitate detection and localization of bleeding sites, and allow faster hemorrhage control. Our eventual goal is to develop a portable automatic HIFU device that can be used at the site of accident to localize and stop the bleeding by a non-skilled operator, similar to current usage of defibrillators. |
| **Task Progress:** | We accomplished research tasks proposed to be performed in the Year 1, and described in Specific Aim 1a (Mechanisms of HIFU-induced hemostasis in the presence of Optison). Methods: A HIFU applicator was scanned over the incision in the liver until hemostasis was achieved. Optison solution (0.8, 1.2, or 1.8 ml) was injected into a mesenteric vein, 20 s before the start of HIFU application. Temperature was measured using infrared thermal camera and a thermocouple connected to LabView acquisition system. Broadband noise, an indication of inertial cavitation, was measured using passive cavitation detection. The tension strength was determined as the applied force at which the hemostatic incision started to bleed. For bleeding detection, triangular incisions were made in the posterior surface of liver and kidneys in heparinized rabbits. The transcutaneous punctures in the femoral arteries were also observed. Bleeding detection was performed using Color Doppler ultrasound and harmonic imaging, before and after injection of Optison (~0.1 ml/Kg). Results: Thermal Measurements. The coagulum temperature was independent of Optison concentration: 75.1 +/- 5.2, 78.9 +/- 5.2, 76.7 +/- 4.2, and 72.7 +/- 9.3 deg C at Optison concentrations of 0, 0.8, 1.2 and 1.8 ml, respectively (n=6-24). Cavitation Detection. The broadband noise measured during HIFU sealing of liver incisions was 1.7 +/- 1.0 (relative units) (n=17) without Optison, and 2.9 +/- 2.3 (n=24) with Optison administration. Coagulum Formation. The coagulum formation times were 9, 7, 5 and 4 s at Optison concentrations of 0, 0.8, 1.2 and 1.8 ml, respectively (n=6-24). The statistical significance ($p<0.05$) between control and Optison treatments was achieved. Live Microscopy. No difference was observed in the appearance of coagulum formed with and without Optison administration. Tension strength. The tension strength at which the hemostatic liver incisions started to bleed was 1.2 +/- 0.7 N, 1.2 +/- 0.5 N and 0.8 +/- 0.3 N for HIFU, Optison+HIFU, and electrocautery treatments, respectively (n=6-11). The incisions sealed with HIFU were 35% stronger ($p<0.05$). Histological Analysis. Histological observations |

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| | showed similar appearance of HIFU and HIFU+UCA lesions. Hemolysis. The concentration of free hemoglobin in blood (indication of blood hemolysis) was on average 12 ml/dL after HIFU treatment, and 51 ml/dL after HIFU+Optison treatment (n=6-9). Bleeding Detection. The administration of Optison increased the rate of successful detection (by ~60%) of occult bleedings from solid organ injuries and vessel punctures. Conclusion. The enhancement of HIFU-induced hemostasis in the presence of Optison appeared to be due to mechanical mechanisms. The presence of UCA resulted in both better visualization of bleeding sites and faster formation of the coagulum over the liver incisions during acoustic hemostasis. |
| **Bibliography Type:** | Description: (Last Updated: 09/06/2019) |
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| **Awards** | N/A "Asked to be a reviewer for Cell and Mol Biol, J Pharm Pharmacol, Pharmacol Res, and Medical Science Monitor International Reviewers Panel" Jan-2004 |
| **Awards** | N/A "Award: 5% salary increase" Jan-2004 |
| **Awards** | N/A "Chaired a session on Therapy Monitoring at IEEE International Ultrasonics Symposium, September 2005 Rotterdam" Jan-2004 |
| **Books/Book Chapters** | Vaezy S, Zderic V, Fujimoto V "Uterine Fibroids: Pathogenesis & Management. Treatment of Uterine Fibroids: New Non-Invasive Developments High Intensity Focused Ultrasound" Ed. Brosens I Taylor&Francis Group, Nov-2005 |
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| **Patents** | 60/610,456 is converting Patent Jan-2006 Vaezy S, Held RT, Zderic V "Interference-Free Ultrasound Imaging During HIFU Therapy, Using Software Tools" |
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| **Patents** | 60/610,451 is converting Patent Jan-2006 Vaezy S, Nguyen TN, Zderic V, Foley JL "Acoustic Coupling using an Independent Water Pillow with Circulation for Cooling the Transducer" |
| **Presentation** | Zderic V, Vaezy S, Crum LA "HIFU Hemostasis of Liver Injuries Enhanced by Ultrasound Contrast Agents" N/A<br>Jan-2005 |
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| **Presentation** | Burgess S, Zderic V, Vaezy S "Image-guided Acoustic Hemostasis" N/A<br>May-2005 |
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| **Presentation** | Vaezy S, Zderic V, Foley J, Luo W, Starr F, Lebedev A "Thermal Effects of HIFU at Soft Tissue-Air Interface in the Post-Focal Region: A Safety Concern, and a Potential Solution using Cavitation" N/A<br>Sep-2005 |
| **Presentation** | Zderic V, Burgess S, Vaezy S "Thin-Profile HIFU Applicators for Intraoperative Hemostasis" N/A<br>Sep-2005 |
| **Presentation** | Zderic V, Crum L, Vaezy S "Selective Liver Vessel Occlusion with an Ultrasound-Guided HIFU Phased Array" N/A<br>Sep-2005 |
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| **Presentation** | Zderic V, Crum LA, Vaezy S "Remote Acoustic Hemostasis (RAH) - Preclinical Studies" N/A<br>Jun-2005 |

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| **Presentation** | Zderic V, Foley J, O Keefe G, Crum L, Vaezy S "High-Throughput HIFU Treatment for Intraoperative Resection of Solid Organs" N/A<br>Sep-2005 |
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